

Operator Notation

		$\mathbf{x}, \mathbf{y} \in \mathbb{C}^N$	$\mathbf{x}, \mathbf{y} \in \ell_2(\mathbb{Z})$	$\mathbf{x}, \mathbf{y}: \mathbb{R} \rightarrow \mathbb{C}$
shift	$\mathbf{y} = \mathcal{S}^d \mathbf{x}$	$y[n] = x[(n+d) \bmod N]$	$y[n] = x[n+d]$	$y(t) = x(t+d)$
modulation	$\mathbf{y} = \mathcal{M}^\varphi \mathbf{x}$	$y[n] = e^{j\varphi n} x[n]$	$y[n] = e^{j\varphi n} x[n]$	$y(t) = e^{j\varphi t} x(t)$
time reversal	$\mathbf{y} = \mathcal{R} \mathbf{x}$	$y[n] = x[-n \bmod N]$	$y[n] = x[-n]$	$y(t) = x(-t)$

Discrete Fourier Transform (DFT)

all signals are finite-length signals: $\mathbf{x}, \mathbf{X} \in \mathbb{C}^N$, $n, k = 0, 1, 2, \dots, N-1$

fundamental frequency:	$\omega_N = \frac{2\pi}{N}$	
DFT basis vectors:	$\{\mathbf{w}_k\}_{n=0,1,\dots,N-1}$	$w_k[n] = e^{j\omega_N kn} = e^{j\frac{2\pi}{N} kn}$
DFT matrix:	$\mathbf{W} \in \mathbb{C}^{N \times N}$	$W[n, k] = e^{-j\frac{2\pi}{N} nk}$
analysis formula:	$\mathbf{X} = \text{DFT}\{\mathbf{x}\} = \mathbf{W}\mathbf{x}$	$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N} nk}$
synthesis formula:	$\mathbf{x} = \text{IDFT}\{\mathbf{x}\} = \frac{1}{N} \mathbf{W}^H \mathbf{X}$	$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N} nk}$
energy conservation:	$\ \mathbf{x}\ ^2 = \frac{1}{N} \ \mathbf{X}\ ^2$	$\sum_{n=0}^{N-1} x[n] ^2 = \frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$
time reversal:	$\mathcal{R} \mathbf{x} \xleftrightarrow{\text{DFT}} \mathcal{R} \mathbf{X}$	$x[-n \bmod N] \xleftrightarrow{\text{DFT}} X[-k \bmod N]$
conjugation:	$\mathbf{x}^* \xleftrightarrow{\text{DFT}} \mathcal{R} \mathbf{X}^*$	$x^*[n] \xleftrightarrow{\text{DFT}} X^*[-k \bmod N]$
time shift:	$\mathcal{S}^d \mathbf{x} \xleftrightarrow{\text{DFT}} \mathcal{M}^{d\omega_N} \mathbf{X}$	$x[(n+d) \bmod N] \xleftrightarrow{\text{DFT}} e^{j\frac{2\pi}{N} dk} X[k]$
frequency shift:	$\mathcal{M}^{m\omega_N} \mathbf{x} \xleftrightarrow{\text{DFT}} \mathcal{S}^{-m} \mathbf{X}$	$e^{j\frac{2\pi}{N} mn} x[n] \xleftrightarrow{\text{DFT}} X[(k-m) \bmod N]$

notable DFT pairs:

$x[n] = \delta[n-d], \quad d \in \mathbb{Z}$	$X[k] = e^{-j\frac{2\pi}{N} dk}$
$x[n] = 1$	$X[k] = N\delta[k]$
$x[n] = e^{j\frac{2\pi}{N} dn}$	$X[k] = N\delta[k-d]$
$x[n] = \cos\left(\frac{2\pi}{N} dn + \phi\right)$	$X[k] = \frac{N}{2} \left(e^{j\phi} \delta[k-d] + e^{-j\phi} \delta[k-N+d] \right)$
$x[n] = \sin\left(\frac{2\pi}{N} dn + \phi\right)$	$X[k] = -j \frac{N}{2} \left(e^{j\phi} \delta[k-d] - e^{-j\phi} \delta[k-N+d] \right)$
$x[n] = \begin{cases} 1 & n < M \\ 0 & M \leq n < N \end{cases}$	$X[k] = \frac{\sin((\pi/N)Mk)}{\sin((\pi/N)k)} e^{-j\frac{\pi}{N}(M-1)k}$

Discrete-Time Fourier Transform (DTFT)

signals are infinite-length sequences, transforms are 2π -periodic complex-valued functions of a real variable

analysis formula:	$\mathbf{X} = \text{DTFT}\{\mathbf{x}\}$	$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$
synthesis formula:	$\mathbf{x} = \text{IDTFT}\{\mathbf{X}\}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega)e^{j\omega n} d\omega$
energy conservation:	$\ \mathbf{x}\ ^2 = \ \mathbf{X}\ ^2$	$\sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) ^2 d\omega$
time reversal:	$\mathcal{R}\mathbf{x} \xleftrightarrow{\text{DTFT}} \mathcal{R}\mathbf{X}$	$x[-n] \xleftrightarrow{\text{DTFT}} X(-\omega)$
conjugation:	$\mathbf{x}^* \xleftrightarrow{\text{DTFT}} \mathcal{R}\mathbf{X}^*$	$x^*[n] \xleftrightarrow{\text{DTFT}} X^*(-\omega)$
time shift:	$\mathcal{S}^d\mathbf{x} \xleftrightarrow{\text{DFT}} \mathcal{M}^d\mathbf{X}$	$x[n+d] \xleftrightarrow{\text{DTFT}} e^{j\omega d} X(\omega)$
frequency shift:	$\mathcal{M}^\varphi\mathbf{x} \xleftrightarrow{\text{DFT}} \mathcal{S}^{\varphi-\varphi}\mathbf{X}$	$e^{j\varphi n} x[n] \xleftrightarrow{\text{DTFT}} X(\omega-\varphi)$
convolution:	$\mathbf{x} * \mathbf{y} \xleftrightarrow{\text{DFT}} \mathbf{X}\mathbf{Y}$	$\sum_k x[k]y[n-k] \xleftrightarrow{\text{DTFT}} X(\omega)Y(\omega)$
modulation:	$\mathbf{xy} \xleftrightarrow{\text{DFT}} \mathbf{X} * \mathbf{Y}$	$x[n]y[n] \xleftrightarrow{\text{DTFT}} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta)Y(\omega-\theta)d\theta$

notable DTFT pairs:

$x[n] = \delta[n-d], \quad d \in \mathbb{Z}$	$X(\omega) = e^{-j\omega d}$
$x[n] = a^n u[n] \quad a < 1$	$X(\omega) = \frac{1}{1 - ae^{-j\omega}}$
$x[n] = \begin{cases} 1 & 0 \leq n < M \\ 0 & \text{otherwise} \end{cases}$	$X(\omega) = \frac{\sin((M/2)\omega)}{\sin(\omega/2)} e^{-j\frac{M-1}{2}\omega}$
$x[n] = \frac{\omega_c}{\pi} \text{sinc}\left(\frac{\omega_c}{\pi} n\right)$	$X(\omega) = \text{rect}\left(\frac{\omega}{2\omega_c}\right)$
$x[n] = 1$	$X(\omega) = \tilde{\delta}(\omega)$
$x[n] = e^{j\omega_0 n}$	$X(\omega) = \tilde{\delta}(\omega - \omega_0)$
$x[n] = \cos(\omega_0 n + \phi)$	$X(\omega) = (1/2)[e^{j\phi}\tilde{\delta}(\omega - \omega_0) + e^{-j\phi}\tilde{\delta}(\omega + \omega_0)]$
$x[n] = \sin(\omega_0 n + \phi)$	$X(\omega) = (-j/2)[e^{j\phi}\tilde{\delta}(\omega - \omega_0) - e^{-j\phi}\tilde{\delta}(\omega + \omega_0)]$

$$\tilde{\delta}(\omega) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k) \text{ (periodized Dirac delta),} \quad \text{sinc}(t) = \frac{\sin \pi t}{\pi t}, \quad \text{rect}(t) = \begin{cases} 1 & |t| < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

Continuos-Time Fourier Transform (CTFT)

signals and transforms are complex-valued functions of a real-valued variable

analysis formula:	$\mathbf{X} = \text{CTFT}\{\mathbf{x}\}$	$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$
synthesis formula:	$\mathbf{x} = \text{ICTFT}\{\mathbf{X}\}$	$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$
energy conservation:	$\ \mathbf{x}\ ^2 = \ \mathbf{X}\ ^2$	$\int_{-\infty}^{\infty} x(t) ^2 dt = \int_{-\infty}^{\infty} X(f) ^2 df$
time reversal:	$\mathcal{R}\mathbf{x} \xleftrightarrow{\text{CTFT}} \mathcal{R}\mathbf{X}$	$x(-t) \xleftrightarrow{\text{CTFT}} X(-f)$
conjugation:	$\mathbf{x}^* \xleftrightarrow{\text{CTFT}} \mathcal{R}\mathbf{X}^*$	$x^*(t) \xleftrightarrow{\text{CTFT}} X^*(-f)$
time shift:	$\mathcal{S}^\tau \mathbf{x} \xleftrightarrow{\text{CTFT}} \mathcal{M}^{2\pi\tau} \mathbf{X}$	$x(t + \tau) \xleftrightarrow{\text{CTFT}} e^{j2\pi\tau f} X(f)$
frequency shift:	$\mathcal{M}^{2\pi\theta} \mathbf{x} \xleftrightarrow{\text{CTFT}} \mathcal{S}^{-\theta} \mathbf{X}$	$e^{j2\pi\theta t} x(t) \xleftrightarrow{\text{CTFT}} X(f - \theta)$
scaling:		$x(at) \xleftrightarrow{\text{CTFT}} \frac{1}{ a } X\left(\frac{f}{a}\right)$
convolution:	$\mathbf{x} * \mathbf{y} \xleftrightarrow{\text{DFT}} \mathbf{X}\mathbf{Y}$	$\int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau \xleftrightarrow{\text{CTFT}} X(f) Y(f)$
modulation:	$\mathbf{xy} \xleftrightarrow{\text{DFT}} \mathbf{X} * \mathbf{Y}$	$x(t) y(t) \xleftrightarrow{\text{CTFT}} \int_{-\infty}^{\infty} X(\theta) Y(f - \theta) d\theta$

notable CTFT pairs:

$x(t) = e^{-at} u(t), \text{Re}\{a\} > 0$	$X(f) = \frac{1}{a + j2\pi f}$
$x(t) = \text{rect}(t)$	$X(f) = \text{sinc}(f)$
$x(t) = \text{sinc}(t)$	$X(f) = \text{rect}(f)$
<i>(all the following deltas are Dirac deltas)</i>	
$x(t) = \delta(t)$	$X(f) = 1$
$x(t) = \delta(t - t_0)$	$X(f) = e^{-j2\pi f t_0}$
$x(t) = 1$	$X(f) = \delta(f)$
$x(t) = e^{j2\pi f_0 t}$	$X(f) = \delta(f - f_0)$
$x(t) = \cos(2\pi f_0 t)$	$X(f) = \frac{1}{2}(\delta(f - f_0) + \delta(f + f_0))$
$x(t) = \sin(2\pi f_0 t)$	$X(f) = \frac{1}{2j}(\delta(f - f_0) - \delta(f + f_0))$

Z-Transform

		ROC
definition:	$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$	R_x
linearity:	$a\mathbf{x} + b\mathbf{y} \xleftrightarrow{\mathcal{Z}} aX(z) + bY(z)$	$R_x \cap R_y$
time shift:	$\mathcal{S}^d \mathbf{x} \xleftrightarrow{\mathcal{Z}} z^d X(z)$	R_x
convolution:	$\mathbf{x} * \mathbf{y} \xleftrightarrow{\mathcal{Z}} X(z) Y(z)$	$R_x \cap R_y$
time reversal:	$\mathcal{R}\mathbf{x} \xleftrightarrow{\mathcal{Z}} X(z^{-1})$	$z z^{-1} \in R_x$
conjugation:	$\mathbf{x}^* \xleftrightarrow{\mathcal{Z}} X^*(z^*)$	$z z^* \in R_x$

notable z-transform pairs:

		ROC
$x[n] = \delta[n - d], \quad d \in \mathbb{Z}$	$X(z) = z^{-d}$	$\mathbb{C}$
$x[n] = \begin{cases} a^n & 0 \leq n < M \\ 0 & \text{otherwise} \end{cases}$	$X(z) = \frac{1 - a^M z^{-M}}{1 - a z^{-1}}$	$\mathbb{C}$
$x[n] = a^n u[n]$	$X(z) = \frac{1}{1 - a z^{-1}}$	$ z > a $
$x[n] = n a^n u[n]$	$X(z) = \frac{a z^{-1}}{(1 - a z^{-1})^2}$	$ z > a $

Multirate

upsampling:	$(\mathcal{U}_N \mathbf{x})[n] = \begin{cases} x[n/N] & n \bmod N = 0 \\ 0 & \text{otherwise} \end{cases}$	$X_{UN}(z) = X(z^N)$
downsampling:	$(\mathcal{D}_M \mathbf{x})[n] = x[Mn]$	$X_{DM}(\omega) = \frac{1}{M} \sum_{k=0}^{N-1} X\left(\frac{\omega - 2\pi k}{M}\right)$
interpolation:	upsampling by N followed by lowpass filtering with cutoff $\omega_c = \pi/N$ and gain N	
decimation:	lowpass filtering with cutoff $\omega_c = \pi/M$ and unit gain, followed by downsampling	

Stochastic and Adaptive

correlation:	$r_{xy}[k] = \sum_{n=-\infty}^{\infty} x^*[n]y[n+k]$ $r_{xy}[k] = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x^*[n]y[n+k]$ $r_{xy}[k] = \mathbb{E}[x^*[n]y[n+k]]$	$\mathbf{r}_{xy} = \mathcal{R} \mathbf{x}^* * \mathbf{y}$ $\mathbf{r}_x = \mathcal{R} \mathbf{x}^* * \mathbf{x}$
Power Spectral Density (PSD):	$P_x(\omega) = \text{DTFT}\{r_x[k]\}$	$\mathbf{r}_x \xleftrightarrow{\text{DTFT}} \mathbf{P}_x$
total power:	$W_x = r_x[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_x(\omega) d\omega$	$W_x = \ \mathbf{x}\ ^2 = r_x[0]$
	PSD of filtered process:	$\mathbf{y} = \mathbf{h} * \mathbf{x} \Rightarrow P_y(\omega) = H(\omega) ^2 P_x(\omega)$
	PSD of white noise:	$P_\eta(\omega) = \sigma_\eta^2 \text{ (constant)}$
	PSD of quantization error, R bps:	$P_e(\omega) = \Delta^2/12, \quad \Delta = (x_{\max} - x_{\min})/2^R$
	SNR of quantized signal, R bps:	$\text{SNR} = 6R \text{ dB}$